

Characterization of lower limb activity during gait using wearable, multi-channel surface EMG and IMU sensors

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Abstract— Lower limb amputees who wear a conventional passive prosthesis usually consume more metabolic energy than normal persons and may put more pressure on their knees, which eventually results in knee damage. In order to model the muscle activity during human gait, a simulated amputee walking on normal subjects was investigated by adding a 10 lbs. load on their lower right limb during level walking and walking on a slope of 10 degrees. Multi-channel surface electromyogram (sEMG) signals from 8 muscle groups were recorded and modeled. Additionally, acceleration on different types of walking was recorded from three positions on the lower body. Several basic features extracted from EMG data were explored. Results shows that the human gait associates with a complex collaboration between various muscles and the effect of load on gait was clearly revealed by the EMG signals recorded from the lower limb muscles. This study was implemented to provide an initial assessment for the optimization of active biomechanical lower limb prostheses.

Keywords—*Prosthetics; electromyography; Tuning; energy expenditure; minimization*

I. INTRODUCTION

Human gait analysis plays an important role in understanding the biomechanical processes associated with gait. Efforts have been made to create effective models for diagnosis and prediction of gait consequences in patients who suffer from motor diseases such as Parkinson's, multiple sclerosis (MS), and prosthetic limb control. Accurate modeling of human gait is required for the purpose of detection, prediction, or assistive control for patients with different clinical conditions. In this study, we focus on the parametric gait modeling of lower limb amputees.

Amputation is a surgical procedure to remove a portion of or an entire limb. The most common reason for amputation is problems with blood flow which occur when arteries become narrow or damaged. There are nearly 2 million people living with limb loss in the United States. Among those living with limb loss, the main causes are vascular disease (54%) – including diabetes and peripheral arterial disease – trauma

(45%) and cancer (less than 2%) [1]. Approximately 185,000 amputations occur in the United States each year [2].

Studies illustrate that amputees need to consume more metabolic energy to keep their balance and perform gait [3]. The more energy consumption as well as excessive pressure applied to the intact limb, in the long run, will have health detrimental consequences that may also change body biomechanics. Furthermore, having an amputation for a long period of time is associated with secondary physical conditions including osteoarthritis, osteoporosis, back pain, and other musculoskeletal problems [4]. The current clinical prosthetic services for prosthesis fitting and tuning does not suffice the goal of maximally reducing the user's energy expenditure. In practice, trained prosthetists conduct the prosthesis fitting and optimization; they tune the powered prostheses based on their observations and experience, as well as user's subjective feedback [5]. In order to overcome these problems, it is important to have an accurate model for changes in gait parameters. Also, having a good understanding of the gait kinematics will result in straightforward and optimum manipulation of powered prosthetics and reduce energy expenditure and pressure on the knee.

Gait analysis is implemented for the purpose to understand the kinematics and kinetics of locomotion. In this study, we focused on the changes in gait while subjects have a load on their lower limb to simulate amputees. The proposed method is enhanced with multi-channel EMG recordings, and multi-channel inertial measurement units (IMU).

In this study, the gait activity is measured by using kinetic sensors and physiological sensors placed on lower limbs. Both normal gait and simulated prosthetic walking using a 10 lbs load was investigated and the data recorded from the devices were modeled. A 10 lbs sandbag was attached to their right lower limb for simulation.

The rest of paper is organized as follows: in section II data acquisition technique is described, section III presents the method employed on data modeling, section IV shows the

results acquired from the data analysis, and in section V discussion regarding the results is provided.

II. EXPERIMENTS AND DATA ACQUISITION

An essential part of this study was to collect a reliable dataset to simulate the effects of gait on the amputees, and to collect comprehensive information regarding their gait. In order to reach this goal, an eight-camera motion tracking system was used in the laboratory to monitor parameter variations during gait while walking on a treadmill. Using this set up, multi-channel electromyography (EMG) signal, inertial measurement unit (IMU) signals were collected and the motion of the subject was observed using a motion tracking system.

A. Electromyography (EMG)

Electromyography (EMG) is a standard clinical tool for recording and analysis of myoelectric signals associated with muscular activity in which the myoelectric signals are resulted from physiological variations in the state of muscle fiber membranes [6].

The concentration of this study was on the lower body muscles and the EMG data were recorded from the following muscles on healthy participants: Soleus, Gastrocnemius Medialis (GM), Gastrocnemius Lateralis (GL), Tibialis Anterior (TA), Vastus Medialis (VL), and Semitendinosus. Shimmer3 EMG wireless EMG device [7] was used for the recording and the sampling rate was set at 1000 Hz. Disposal silver/silver chloride surface electrode pads were placed on the surface of the above muscle groups, with respect to the sensor locations recommended by SENIAM [8].

B. Inertial Measurement Units (IMU)

Each of the Shimmer3 devices is embedded with an IMU including three channels of accelerometer and three channels of gyroscope. The IMU data was also sampled at 1000 Hz and recorded with EMG signal.

C. Database

In order to obtain data, three male subjects with normal gait activity were investigated (age of 24.0 ± 2.0 , and BMI of 21.3 ± 1.3) and EMG signals from the muscles that were mentioned above was recorded. Alongside the EMG signal, accelerometer data was recorded from three parts on the body as well. The data was obtained from subjects while they walked on a treadmill at a preset speed for durations of three minutes. The recording procedure was done in four different states of walking. These states were; walking on a level surface with and without load on the right lower limb, as well as, walking with and without load on an elevated plane raised by ten degrees. To enable a load on the limb which was considered as amputee a 10 lbs. sandbag was used, which was attached to the right ankle. The sandbag was used to simulate the amputee condition and observe the effect of the load on the parameters of the gait to comprehend which muscles and what degree of activity is present during each individual gait cycle for each of the four walking states.

III. METHODS

The recorded EMG and IMU data in this study were analyzed statistically to explore the complex activation pattern of muscles during gait and to understand which muscles and with what degree participate in this action.

A. pre-processing and artifact removal

EMG signal may be contaminated by large motion artifact due to the movement between the electrode and its contact to the body. To eliminate the motion artifacts in EMG signal a Butterworth high pass filter with the cut off frequency of 30Hz was employed. After baseline drift was corrected, the EMG signal was rectified to obtain positive amplitudes. In order to eliminate the steep amplitude spikes in the EMG signal, the signal was smoothed using a moving average filter with a window size of 100 millisecond. Fig. 1 shows the high pass filtered EMG signal for the Gastrocnemius Medialis muscle in four states of walking for 2 seconds. The accelerometer data was also smoothed in each axis in a window of 100 millisecond.

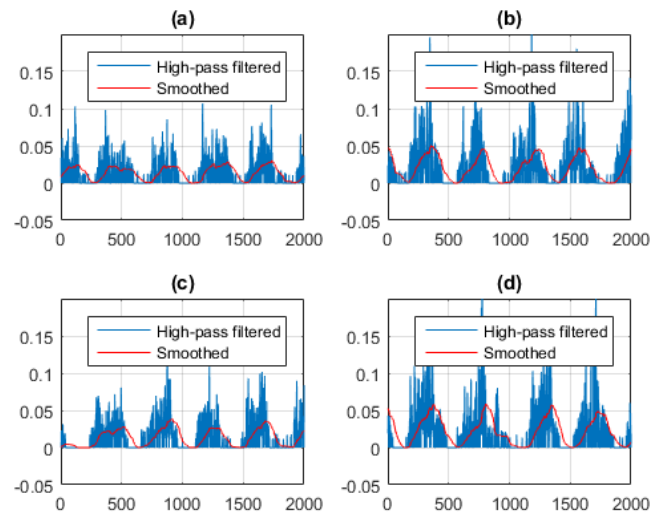


Figure 1. Electrical activity of Gastrocnemius Medialis muscle. (a) EMG recorded in level walking without load. (b) EMG signal in level walking with 10lbs. load on the lower right leg. (c) EMG signal in walking up a 10 degree slope without load. (d) EMG signal during walking up 10 degree slope and with load on the leg.

B. Feature Evaluation

A set of statistical features was obtained from the EMG signal during this study. The features used here were calculated from ten episodes of the recorded data. The features used for EMG signal are listed below:

- Mean
- Variance
- Skewness
- Kurtosis
- Standard deviation
- Power of the signal
- Root mean square
- Mean frequency

In order to reduce the complexity of the proposed method, each feature was analyzed for one of the subjects to determine the best single feature. One-way analysis of variance was done for each feature separately for four different states of walking and the p-values were calculated for them. From the mean p-values that have been calculated, standard deviation feature depicted the smallest p-value (4.49e-02). So for further analysis standard deviation was chosen to investigate its distribution in walking with different loads and slopes.

The IMU sensor used in this study was implemented to analyze the acceleration the movement of the left and right hip surfaces, as well as the lower left limb at the shin. For analysis of the accelerometer data three features were calculated from the recording. The features used here are: mean speed, mean radius, and mean frequency. In order to calculate these features the total distance covered in time t is defined as follows:

$$D_t = \sum_{i=0}^{N-1} \sqrt{(d_{y_i} - d_{y_{i+1}})^2 + (d_{x_i} - d_{x_{i+1}})^2} \quad (1)$$

where, N is the number of samples in time duration of t . The mean speed, mean radius, and mean frequency are respectively defined as:

$$s_m = \frac{D_t}{t} \quad (2)$$

$$r_m = \frac{1}{N} \sum_{i=0}^{N-1} \sqrt{d_{x_i}^2 + d_{y_i}^2} \quad (3)$$

$$f_m = \frac{D_t}{2\pi r_m t} \quad (4)$$

These accelerometer features calculated from the IMU recordings.

IV. RESULTS

Using the features calculated in the previous section the variation in the gait parameters measured were investigated, these results were then used to determine which measurements provided the most useful physiological indicators which could be used in parameter optimization for active lower limb prosthetics. As mentioned in the previous section, surface EMG signals were recorded from eight muscles located on thighs and left lower limb. Each recording was divided into 10 segments and the standard deviation for smoothed version of each segment calculated in the different states of walking. The distribution for this feature in different walking states were compared to the initial state of walking without a load and without increment for each subject. As expected, to optimize the prosthetic limb for each subject, individual comparisons for each subject were also performed. Results of variance analysis for the subjects are depicted in Table I.

Furthermore, the variation in accelerometer data was observed via three features that we defined. These features also derived from segments of the recordings after they have been divided into 10 parts. Results for these data did not provide a

good differentiation measure between walking with load, and for the investigated subjects the physical movement accelerations did not present a meaningful variation, which shows the physiological adaptation of gait parameters in different states.

V. DISCUSSION

Variations in different muscle group activation and acceleration of lower body sections were analyzed, this study was performed to comprehend and detect the parameters which are affected by load addition and walking on an elevated slope.

The variation analysis for EMG signals recorded from 8 different muscles on the lower body showed that most of the muscles collaborate in the complex procedure of gait, and the variation in their standard deviation illustrated a meaningful difference for most of the muscles. However, for some muscles in specific subjects the variation was not significant enough. For example, as it can be seen from the Table I, the Semitendinosus muscle for right leg of subject 1 and 3 did not present a meaningful difference in distribution for the case of walking up a slope with 10 degrees (p-value>0.05). However, for the second subject the same muscle resulted in meaningful difference. Beginning with the high complexity of gait activity in individual muscles, which varies from person to person, and even from gait cycle to gait cycle, in addition to the fact that each individual has a different physiological adaptation to the added load. Followed by the fact that, EMG signal is highly sensitive to noise, and the environment in the lab contains various noise sources which are not easily filtered with high-pass filters. A more accurate filter is necessary for EMG analysis.

Even though some minimal differences have been observed in each subjects, the overall discriminatory effect was good for most of the muscles. Also, it was observed that for the Soleus, ST for left thigh, and VM for left thigh depicted completely different distribution for all the states of walking. This suggests that the activity in these three muscles, for all subjects, increased with the adaptation of load and slope intensification in order to sustain gait.

On the other side, the analysis of the accelerations on both sides of the hip and the acceleration in the lower left limb did not show a meaningful difference between the various states of walking which have been investigated. This suggests that the muscles bear most of the compensation for gait with increase of load in the lower limb. This was consistent with the visual inspection gait, which did not illustrate a significant difference in body movement while bearing a load.

In future works, different muscle groups will be added to the database to further analyze muscle activity, and specific reliance during gait, we will also include wider variability in BMI, age and sex. Which will allow for a more comprehensive system. The results will be used to optimize a bio-electrically controlled prosthetic limb, to help lower limb

amputees to adapt various everyday tasks effortlessly, without extra metabolic energy expenditure and automatically detect the intension from the muscle activity and change the parameters of the biomechanical prosthetic limb.

TABLE I. RESULTS OF ONE WAY VARIANCE ANALYSIS FOR LEVEL WALKING WITHOUT LOAD TO OTHER FOR STATES OF WALKING WITH AND WITHOUT LOAD WHILE WALKING ON NO SLOPE AND, 10 DEGREE SLOPE

Muscle	10 lbs., without slope			0 lbs., with 10 degree slope			10 lbs., with 10 degree slope		
	Subject 1	Subject 2	Subject 3	Subject 1	Subject 2	Subject 3	Subject 1	Subject 2	Subject 2
Soleus	2.42E-07	8.75E-04	3.02E-02	1.92E-12	8.47E-17	1.85E-05	2.16E-11	1.04E-13	1.04E-04
Left ST	0.03335	3.46E-03	1.23E-05	0.0004	3.03E-09	8.85E-14	5.6E-07	1.37E-09	2.72E-14
Right ST	0.000117	3.05E-03	0.681782*	0.381*	1.31E-05	0.42*	1.9E-10	2.51E-03	3.91E-04
GM	3.13E-09	4.50E-03	0.962422*	6.84E-09	1.09E-06	0.2*	0.0002	4.32E-10	3.71E-05
Left VM	1.37E-05	3.77E-05	7.54E-01	2.157E-06	1.53E-11	6.77E-03	0.0047	1.17E-11	2.13E-06
Right VM	0.015417	3.45E-24	2.91E-02	0.016*	3.50E-24	8.00E-04	0.017	3.84E-21	0.365*
TA	0.417456*	-	0.149675	0.00067	-	4.76E-07	6.52E-07	-	2.82E-06
GL	0.192046*	-	1.73E-04	5.35E-07	-	4.09E-02	1.67E-08	-	0.534*

VI. CONCLUSION

The proposed method showed that human gait consists of a complex collaboration between various muscles. The results show that the variation between walking with a load attached to lower limb could be detected by simple analysis

Results of this study could be extended to automatically control the bio-mechanical prosthetic limb for lower limb amputees.

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